

- 8 (a) (i) The emission of electrons when light of sufficient frequency strikes the photocathode. 1
- (ii) 1 dynode ~ multiply by 3, so 9 dynodes ~ $3^9 = 19683$ 1
- (iii) Let n be number of photoelectrons.
 $I = (n/t)e$ 1
 $7.2 \times 10^{-6} = n/t (1.60 \times 10^{-19})$
 At the ANODE, $n/t = 4.5 \times 10^{13}$ 1
- AT the PHOTOCATHODE,
 $n/t = 4.5 \times 10^{13} / 19683$ 1
 $= 2.286 \times 10^9$ 1
- (iv) $E = hf = hc/\lambda$ 1
 $= (6.63 \times 10^{-34})(3.00 \times 10^8) / (410 \times 10^{-9}) = 4.85 \times 10^{-19} \text{ J}$ 1
- (v) $KE_{\max}$ of electrons emitted at photocathode = $hf - \phi$ 1
 $= (\text{iv}) - 3.5 \times 10^{-19}$
 $= 1.35 \times 10^{-19} \text{ J}$ 1
- (vi) gain in KE = WD by p.d. 1
 $\frac{1}{2} mv^2 - (v) = eV$
 $v^2 = [(1.60 \times 10^{-19}) (100) + 1.35 \times 10^{-19}] / (\frac{1}{2} 9.11 \times 10^{-31})$
 $v = 5.95 \times 10^6 \text{ m s}^{-1}$ 1
- (b) (i) 1. emission line spectrum 1
 2. absorption line spectrum 1
- (ii) 1. $E_3 - E_2 = hf$ 1
 $-1.36 \times 10^{-19} - E_2 = (6.63 \times 10^{-34})(1.60 \times 10^{14})$
 $E_2 = -2.4208 \times 10^{-19}$ 1
- $E_2 - E_1 = hf = hc/\lambda$
 $-2.4208 \times 10^{-19} - (-5.46 \times 10^{-19}) = (6.63 \times 10^{-34})(3.00 \times 10^8) / \lambda$
 $\lambda = 654 \text{ nm}$ 1
2. $E_3 - E_1 = hf = hc/\lambda$ 1
 $-1.36 \times 10^{-19} - -5.46 \times 10^{-19} = (6.63 \times 10^{-34})(3.00 \times 10^8) / \lambda$
 $\lambda = 485 \text{ nm}$ 1
3. blue / indigo / violet / purple 1